

Due Wednesday, October 7, 2026 at 11:59 PM as PDF uploaded via Gradescope. For ease of grading, use this document as your template (either print/write/upload, use L^AT_EX or edit using a tablet device) **Show as much work as possible in order to get FULL credit.** There are 7 problems with a total of 30 points available. **Important:** If you use Python for any probability computations, briefly write/include the statements you used to arrive at your answers. If instead you use probability tables, note this in the respective solution, as well.

Problem 1 (5 points)

Respond “T” (True) or “F” (False) to the following statements. Use the boxes provided. Each response is worth 1 point.

(i)

☐

If Z represents the standard normal variable, then the mean of Z is 1.

(ii)

☐

To standardize a normally distributed random variable, we find the difference from its mean and divide the result by its standard deviation.

(iii)

☐

The lifetime of a lightbulb is normally distributed with $\mu = 1400$ hrs and $\sigma = 200$ hrs. The 20th percentile of the lifetimes is approximately 1140 hrs.

(iv)

☐

The lifetime of a lightbulb is normally distributed with $\mu = 1400$ hrs and $\sigma = 200$ hrs. The probability that the lightbulb will last more than 1700 hrs is 0.933.

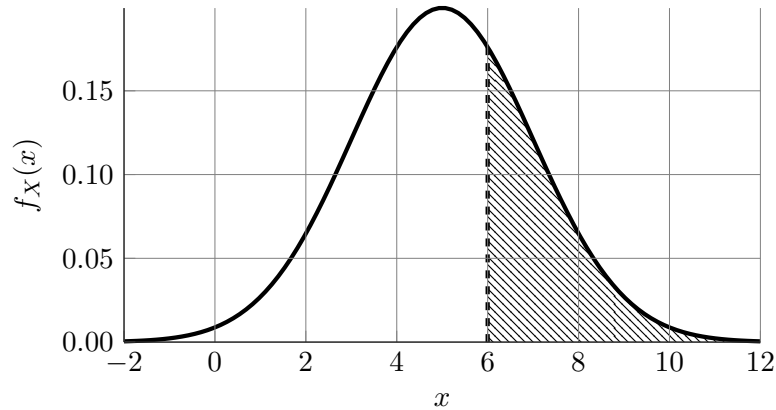
(v)

☐

The area under a PDF can be less than or equal to 1.

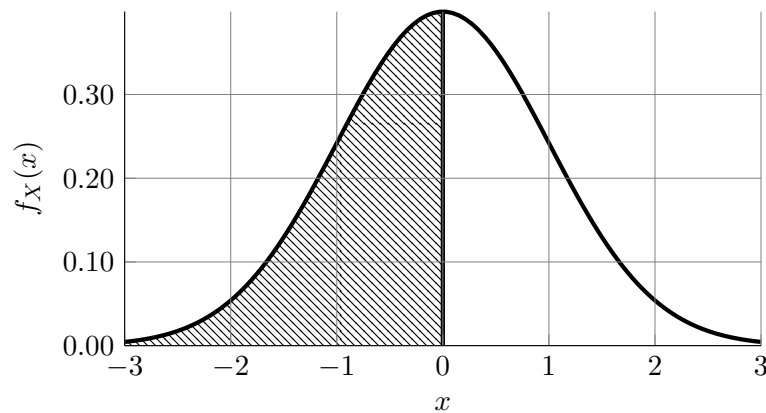
Problem 2 (2 points)

- [1] (a) Write down the expression of the probability represented by the shaded portion of the normal PDF below. For example, $P(X \leq 2)$. Note that a dashed vertical boundary indicates “>” or “<,” while a solid vertical boundary indicates “≥” or “≤.”



Answer:

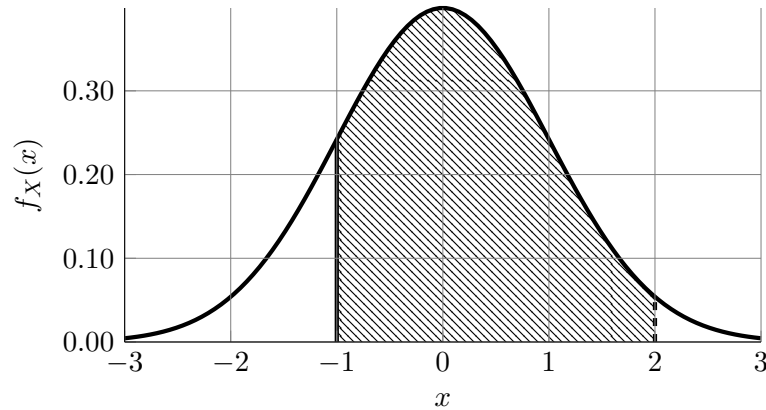
- [1] (b) Write down the expression of the probability represented by the shaded portion of the normal PDF below. For example, $P(X \leq 2)$. Note that a dashed vertical boundary indicates “>” or “<,” while a solid vertical boundary indicates “≥” or “≤.”



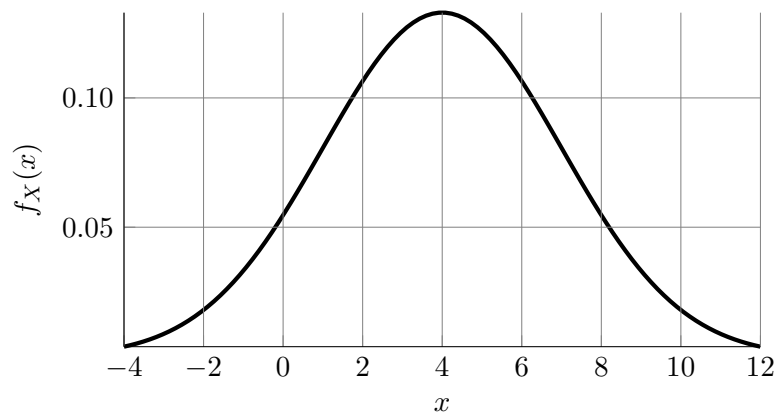
Answer:

Problem 3 (2 points)

- (a) Write down the expression of the probability represented by the shaded portion of the normal PDF below. For example, $P(X \leq 2)$. Note that a dashed vertical boundary indicates “>” or “<,” while a solid vertical boundary indicates “≥” or “≤.” [1]

**Answer:**

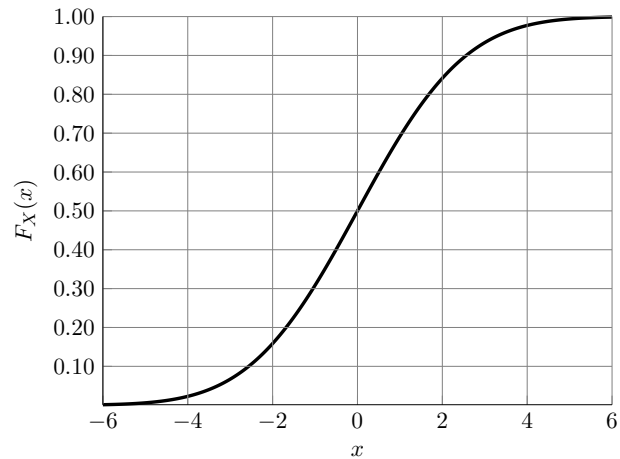
- (b) Below is the PDF of a given normal distribution. What is the median of this distribution? [1]

**Answer:**

Problem 4 (4 points)

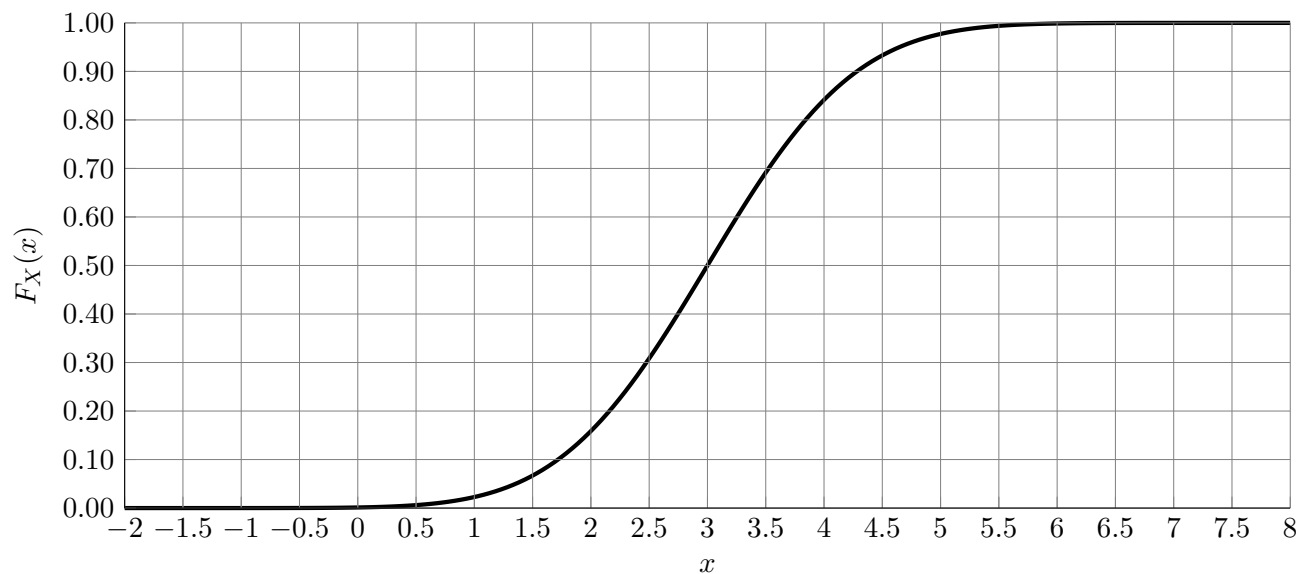
In the following problems, show how you arrive at the answer on the graph.

- [2] (a) Below is the CDF of a given normal distribution. What is the mean of this distribution?



Answer:

- [2] (b) Below is the CDF of a given normal distribution. Estimate the probability $P(X > 3.5)$.



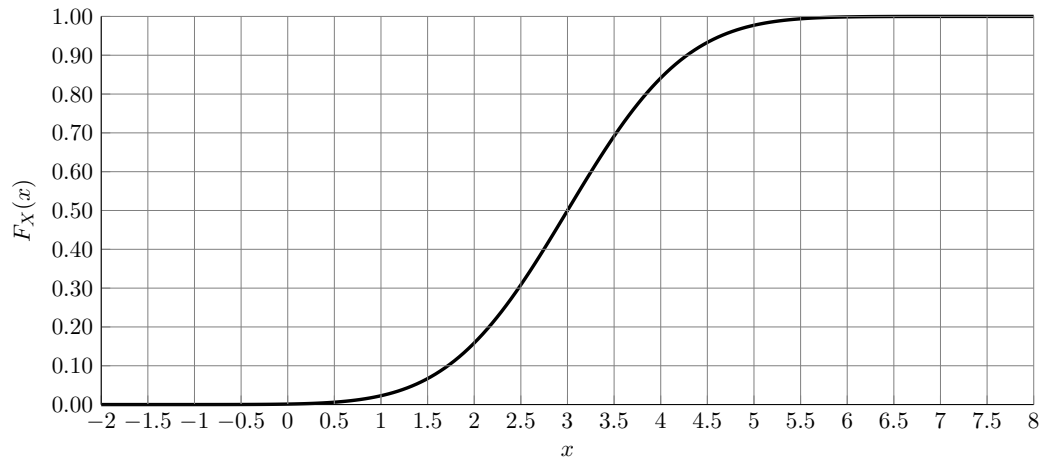
Answer:

Problem 5 (4 points)

In the following problems, show how you arrive at the answer on the graph.

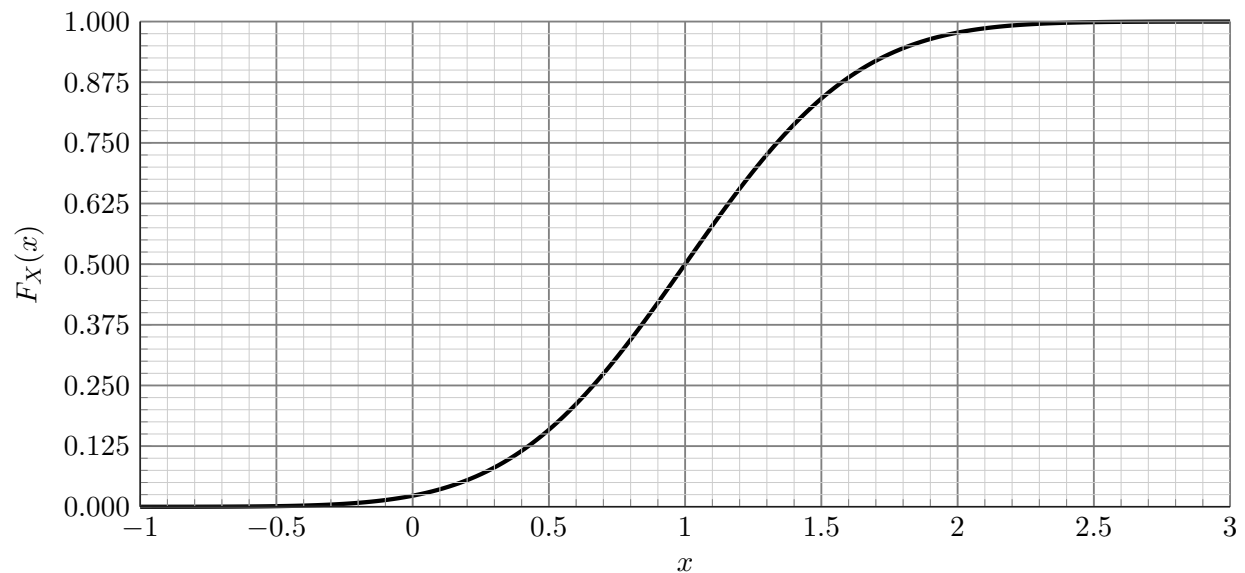
- (a) Below is the CDF of a given normal distribution. Estimate the quantity $F_X^{-1}(0.3)$.

[2]

**Answer:**

- (b) Below is the CDF of a given normal distribution. Estimate the first quartile of the distribution.

[2]

**Answer:**

Problem 6 *Standard Normal Distribution (8 points)*

What percent of a standard normal distribution $\mathcal{N}(\mu = 0, \sigma = 1)$ is found in each region? Sketch the accompanying curve along with your answer.

[2 pts] (a) $Z < -1.35$

[3 pts] (b) $-0.4 < Z < 1.5$

[3 pts] (c) $|Z| > 2$

Problem 7 *Normal Distribution (5 points)*

The average daily high temperature in June in LA is 77°F with a standard deviation of 5°F . Suppose that the temperatures in June closely follow a normal distribution.

- (a) What is the probability of observing an 83°F temperature or higher in LA during a randomly chosen day in June? [2]
- (b) How cool are the coldest 10% of the days (days with lowest average high temperature) during June in LA? [3]